Machine Learning for Early Prediction of Spacecraft Telemetry Anomalies

A graduate project proposed in partial fulfillment of the requirements
for the degree of Master of Science in Computer Science

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George Wang, Committee Chair Date

Objective

Spacecraft are constantly transmitting telemetry data, hundreds of channels of temperatures, voltages, and other system states. This is a lot of data, and it usually results in operators being reactive rather than proactive.

I want to use real satellite data to predict when something will go wrong before the anomaly actually occurs. I will compare various machine learning models and see how early they can reliably predict this, at intervals such as 5, 15, and 30 minutes beforehand. I also want to look at whether some types of anomalies can be predicted earlier than others while keeping false alarms under control.

My main research question is: how far in advance can different spacecraft anomaly classes be predicted from telemetry at a controlled false alarm rate, and how much of the available predictive signal can different machine learning approaches capture? Essentially, I am asking how much warning you can actually get before an anomaly happens on a spacecraft. Does it depend on the type of anomaly? And do better models give you that warning earlier?

Problem Statement

A modern spacecraft continuously generates vast amounts of telemetry data that gets sent back to operators, who use it to monitor the health and status of the spacecraft. Due to the number of satellites in orbit having grown quickly in recent years, the responsibility to maintain and oversee all this data is increasing significantly. As spacecraft systems produce increasingly large and complex volumes of data, manual monitoring of thousands of these channels becomes impractical, increasing the need for

automated anomaly detection methods (Fejjari et al., 2025). Current monitoring systems can detect abnormal behavior and alert operators once something has already started to go wrong, but detecting an anomaly after it begins still leaves them reacting to the issue rather than being proactive.

Machine learning has increasingly been applied to anomaly detection in spacecraft telemetry (Kotowski et al., 2024). What I am interested in is whether the telemetry contains useful warning signs before an anomaly actually starts. There is some evidence this is possible. In the ESA benchmark results, one forecasting-based algorithm detected most anomalies before their annotated start time, but this was treated as a scoring problem rather than something useful (Kotowski et al., 2024).

One complication is that operators do not automatically see early warnings as a positive. The ESA benchmark authors report that detections occurring before an anomaly starts may be treated as false positives by operators if they cannot confirm the anomaly, which can reduce trust in the system (Kotowski et al., 2024). Their timing metric penalizes early detection for this reason. This project takes a different angle by having lead times measured and handling operator trust separately through the false alarm constraint.

Not all spacecraft anomalies behave the same way either. Some may show gradual changes in the telemetry before they occur, while others may happen suddenly with little or no warning. A prediction system also has to avoid producing too many false alarms, since a system that constantly warns operators about anomalies that never happen would not be useful in practice and would likely be ignored.

Technical Approach

I will use the ESA Anomaly Dataset, which is real telemetry from three ESA missions and is an open dataset. I am going to focus on Mission 1. It has 76 channels, 58 of which are actually monitored for anomalies, and it covers about fourteen years. Mission 1 has 200 labeled events in it, and 118 of those are actual anomalies (Kotowski et al., 2024). Each one tells me which channels were affected, when it started and ended, what class it belongs to, and some information about the shape of the anomaly. If time permits, I will add Mission 2 to work with.

The telemetry will first need to be preprocessed so that the different channels can be used together. I will begin with a smaller subset of channels to keep things manageable before scaling up. I may also use or adapt the preprocessing methods provided with the ESA dataset and benchmark rather than rebuilding everything from scratch.

For each prediction horizon, I will take a window of past telemetry and label it positive if an anomaly begins within that period after the window ends. The window always ends before the prediction period starts, so the model never sees the anomaly or the time it is supposed to be predicting. If that data leaks in, I am detecting rather than predicting, so I will verify that this does not happen.

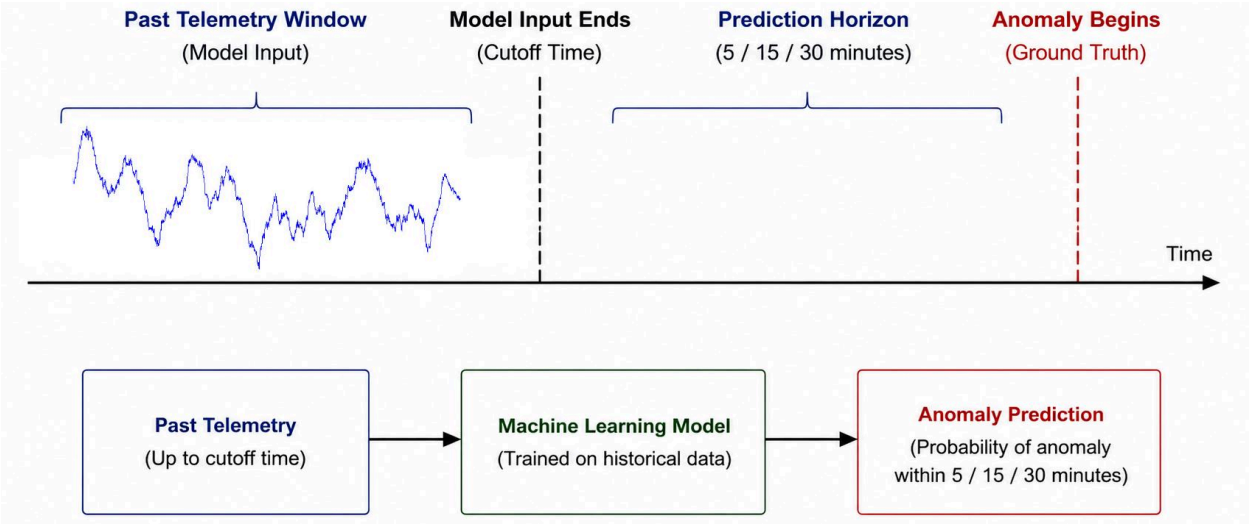


Figure 1. Proposed telemetry window and anomaly prediction framework.

Anomalies are rare compared to normal operation, so I will need to account for class imbalance. I plan to start with simpler machine learning models such as XGBoost or random forest using features from each window, then compare them against one or more temporal models such as an LSTM or 1D CNN. However, with only 118 anomalies in that mission, the simpler models may do just as well, which would be worth reporting.

As mentioned previously, I will test prediction horizons of 5, 15, and 30 minutes. Accuracy alone would be misleading since a model could say nothing is wrong and be right most of the time, so I will use event-level precision and recall along with false alarm rate. Models will be evaluated at comparable false alarm levels so that differences in lead time can be meaningfully compared. I will also split the data by time, training on earlier data and testing on later data, so the model never sees the future.

The risk is that some anomalies may have little or no warning beforehand, but that does not hurt the project since identifying which ones can and cannot be predicted is part of the point. There are also not many anomalies, especially after splitting the data

into training and testing sets. I will limit class-level conclusions to classes with enough events and scale up to more channels if needed once the initial pipeline is working.

The scope of this project is an offline study using historical telemetry. I will not be building a real-time application that runs on the actual spacecraft or fixes anomalies. The goal is simply to measure how much advance warning is obtainable from real telemetry, whether that varies by anomaly class, and how well different models capture it. I will not be determining what warning time operators would actually consider acceptable, since that would require input from spacecraft operations engineers.

Schedule

Period	Task	Output
Prior to work	Identify research area, review existing work, select dataset, proposal development	Research question, project scope, and written proposal
Semester 1 (Sep - Dec 2026)	Explore the ESA dataset and become familiar with the telemetry data	Understanding of the dataset and project requirements
Semester 1 (Sep - Dec 2026)	Prepare the telemetry data and develop the labeling and evaluation	Dataset and pipeline ready for machine learning experiments
Semester 1 (Sep - Dec 2026)	Begin running baseline machine learning models	Initial prediction results
Semester 2 (Jan - May 2027)	Continue model development and compare different ML approaches	Results across models and prediction horizons
Semester 2 (Jan - May 2027)	Analyze results and anomaly detections	Final analysis and conclusions
Semester 2 (Jan - May 2027)	Write thesis and prepare for defense	Completed thesis

Key Performance Indicators (KPI)

- A working system to turn the raw telemetry into labeled windows
- Prediction results at approximately 5, 15, and 30 minutes, reported with false alarm rates and not just accuracy
- Models compared under the same false alarm limit so the lead times mean something
- A breakdown of how much warning is achievable for the anomaly classes that have enough events
- A comparison of the simpler models against the temporal ones, including if the simple ones hold up
- Code and results documented well enough that someone else could reproduce them.

References

De Canio, G., Kotowski, K., & Haskamp, C. (2025). ESA Anomaly Dataset [Dataset]. European Space Agency. https://doi.org/10.5281/zenodo.15237121

Fejjari, A., Delavault, A., Camilleri, R., & Valentino, G. (2025). A Review of Anomaly Detection in Spacecraft Telemetry Data. Applied Sciences, 15(10), 5653. https://doi.org/10.3390/app15105653

Kotowski, K., Haskamp, C., Andrzejewski, J., Ruszczak, B., Nalepa, J., Lakey, D., Collins, P., Kolmas, A., Bartesaghi, M., Martinez-Heras, J., & De Canio, G. (2024). European Space Agency Benchmark for Anomaly Detection in Satellite Telemetry. *ArXiv*. https://arxiv.org/abs/2406.17826